Lecture 16 2025-10-23

3.3. Divided Differences

Goal: Generate a polynomial from values at $x_0, \ldots, x_n$
(v.s. Neville: generate successive approximations to f at a point)

Setup: Let $P_n(x)$ interpolate $f(x)$ at $x_0, \ldots, x_n$ and write it in the form

$$(*) \quad P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + \ldots + a_n(x-x_0)\cdots(x-x_{n-1})$$

Observations:

1) Truncation property: If $k<n$ then

$$(**) \quad P_k(x) = a_0 + a_1(x-x_0) + \ldots + a_k(x-x_0)\cdots(x-x_{k-1})$$

the coefficients don't change when you truncate.

Reason: The remaining terms $a_{k+1}(x-x_0)\cdots(x-x_k), \ldots$
$a_n(x-x_0)\cdots(x-x_n)$ are all zero at $x_0, \ldots, x_k$.
So dropping them reduces the degree of the polynomial to k while preserving the interpolation property at $x_0, \ldots, x_k$, i.e. the polynomial agrees with $f(x_j)$ at $x_0, \ldots, x_k$
Since $P_k(x)$ is the unique polynomial of degree k that agrees with $f(x)$ at $x_0, \ldots, x_k$, the formula on the right hand side of $(**)$ with coefficients $a_0, \ldots, a_k$ from P_n must agree with $P_k(x)$ everywhere

Note: ① implies that adding more terms does not affect the lower degree polynomials.

② If you expand the right hand side of (*) you get

$$a_n x^n + (\text{polynomial of degree } n-1)$$

Definition: (kth divided difference)

We let $f[x_0] = a_0 = f(x_0)$, $f[x_1] = \underbrace{f(x_1)}_{x_1}, \dots, f[x_n] = \underbrace{f(x_n)}_{x_n}$

Then we define recursively

for $i < j$

$$(\Delta) \quad f[x_i, x_{i+1}, \dots, x_{j-1}, x_j] = \frac{f[x_{i+1}, \dots, x_{j-1}, x_j] - f[x_i, \dots, x_{j-1}]}{x_j - x_i}$$

We call $f[x_0, \dots, x_k]$ with $0 \le k \le n$ the **kth divided difference** for $(x_i, f(x_i))$, $i = 0, \dots, n$

Examples: $f[x_i, x_{i+1}] = \dfrac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

$$= \frac{\dfrac{f(x_{i+2}) - f(x_{i+1})}{x_{i+2} - x_{i+1}} - \dfrac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}}{x_{i+2} - x_i}$$

Lemma: For $n \ge 1$ define a_n as the nth coefficient of $P_n(x)$ as in (*) $P_n(x) = a_0 + a_1(x - x_0) + \dots + a_n(x - x_0) \cdots (x - x_{n-1})$

Then

$$a_n = f[x_0, \dots, x_n]$$

Proof: We use induction. We already have $f[x_0] = f(x_0) = a_0$.

So we assume that the claim holds for n points.

Then we have on one hand

$$P_{n-1}(x) = \underbrace{\tilde{P}_{n-2}(x)}_{\text{interpolates at } x_1,\dots,x_{n-1}} + \underbrace{f[x_1,\dots,x_{n-1},x_0]}_{= f[x_0,\dots,x_{n-1}] \text{ Exercise!}}(x-x_1)\cdots(x-x_{n-1})$$

P_{n-1} interpolates at $x_0,\dots x_{n-1}$

on the other hand

$$\tilde{P}_{n-1}(x) = \underbrace{\tilde{P}_{n-2}(x)}_{\substack{\text{interpolates} \\ \text{at } x_1,\dots,x_{n-1}}} + f[x_1,\dots,x_n](x-x_1)\cdots(x-x_{n-1})$$

$\tilde{P}_{n-1}$ interpolates at $x_1,\dots,x_n$

We plug in x_n into P_{n-1} and $\tilde{P}_{n-1}$

$$P_{n-1}(x_n) = \tilde{P}_{n-2}(x_n) + f[x_0,\dots,x_{n-1}](x_n-x_1)\cdots(x_n-x_{n-1})$$
$$\tilde{P}_{n-1}(x_n) = f(x_n) = \tilde{P}_{n-2}(x_n) + f[x_1,\dots,x_n](x_n-x_1)\cdots(x_n-x_{n-1})$$

We also have $P_n(x_n) = f(x_n) = P_{n-1}(x_n) + a_n(x_n-x_0)\cdots(x_n-x_{n-1})$

Plugging in, we get

$$\begin{aligned} f(x_n) &= P_{n-1}(x_n) + a_n(x_n-x_0)\cdots(x_n-x_{n-1}) \\ &= \tilde{P}_{n-1}(x_n) = \tilde{P}_{n-2}(x_n) + f[x_1,\dots,x_n](x_n-x_1)\cdots(x_n-x_{n-1}) \\ &= \tilde{P}_{n-2}(x_n) + f[x_0,\dots,x_{n-1}](x_n-x_1)\cdots(x_n-x_{n-1}) + a_n(x_n-x_0)\cdots(x_n-x_{n-1}) \end{aligned}$$

Solve for a_n:

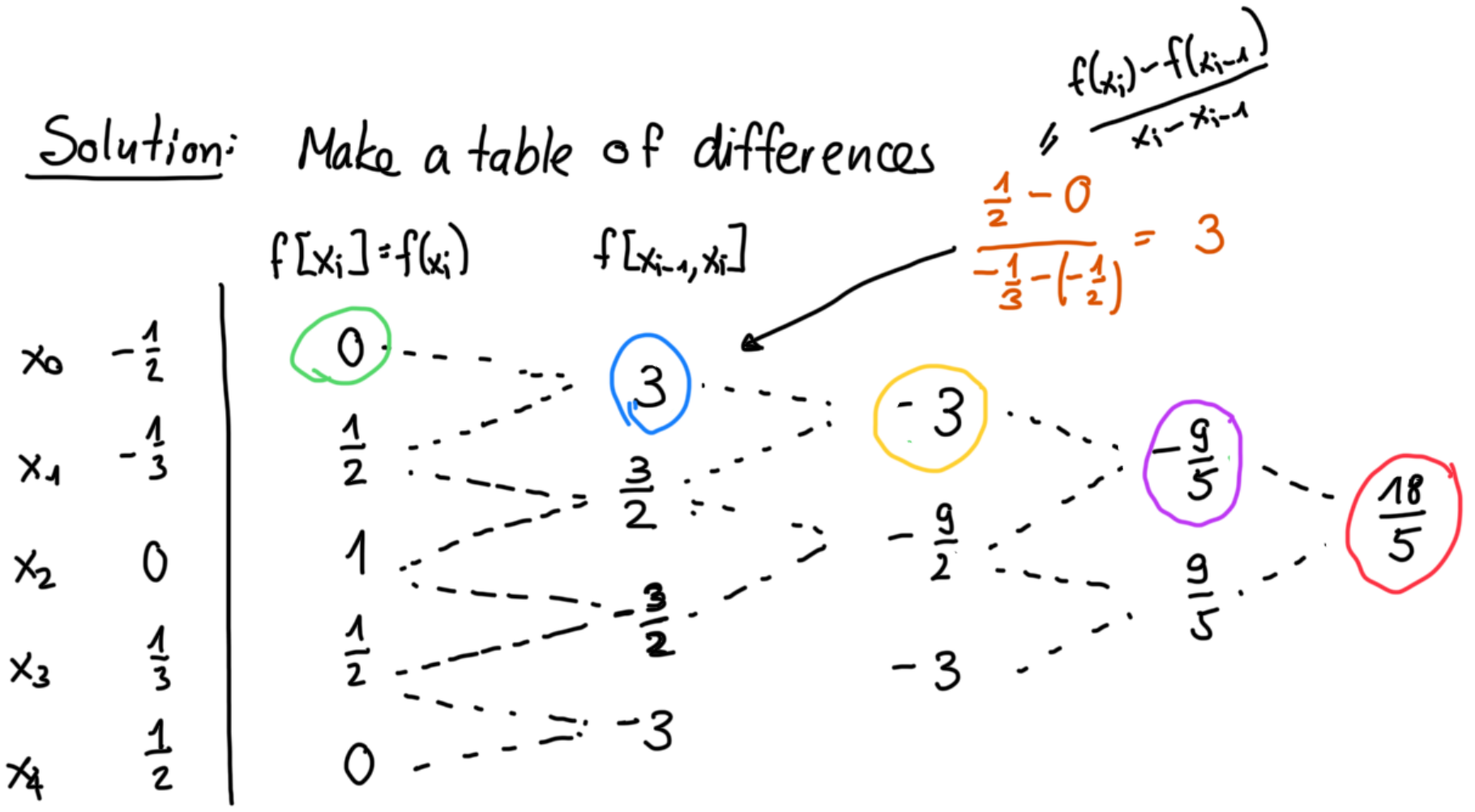
$$f[x_1,\dots,x_n] = f[x_0,\dots,x_{n-1}] + a_n(x_n-x_0)$$

$$\iff a_n = \frac{f[x_1,\dots,x_n] - f[x_0,\dots,x_{n-1}]}{x_n-x_0}$$

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Example: Interpolate $f(x) = \cos(\pi x)$ at

$$x_i = -\frac{1}{2}, -\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{2}$$



$$P_4(x) = 0 + 3\left(x+\tfrac{1}{2}\right) - 3\left(x+\tfrac{1}{2}\right)\left(x+\tfrac{1}{3}\right) - \frac{9}{5}\left(x+\tfrac{1}{2}\right)\left(x+\tfrac{1}{3}\right)x + \frac{18}{5}\left(x+\tfrac{1}{2}\right)\left(x+\tfrac{1}{3}\right)x\left(x-\tfrac{1}{3}\right)$$

↝ can add a new point without affecting previous work

We can follow any consecutive path through the difference table to reconstruct $P(x)$ (add points in various orders)

e.g.

x_0	a			
x_1	.	b		
x_2	.	.	c	d
x_3	.	.	.	

$$P(x) = a + b(x-x_0) + c(x-x_0)(x-x_1) + d(x-x_0)(x-x_1)(x-x_2)$$

x_0	.			
x_1	.	.		
x_2	$\tilde{a}$	.	.	
x_3	.	$\tilde{b}$	$\tilde{c}$	$\tilde{d}$

$$P(x) = \tilde{a} + \tilde{b}(x-x_2) + \tilde{c}(x-x_2)(x-x_3) + \tilde{d}(x-x_2)(x-x_3)(x-x_1)$$

etc.

Stirling's Formula: Average results from two paths through

center of difference table

$$\frac{1}{2}P(x) + \frac{1}{2}P(x) = P(x)$$

(useful if f is odd or even and nodes are symmetric around origin)

x_{-2}					
x_{-1}		b_1		d_1	
x_0	a		c		e
		b_2		d_2	
x_1					
x_2					

$$P(x) = a + \frac{b_1+b_2}{2}(x-x_0) + c(x-x_0)\left(x - \frac{x_{-1}+x_1}{2}\right) + \frac{d_1+d_2}{2}(x-x_0)(x-x_1)(x-x_{-1}) + e(x-x_0)(x-x_{-1})(x-x_1)\left(x - \frac{x_2+x_{-2}}{2}\right)$$

Example from earlier: $f(x) = \cos(\pi x)$

x_0	$-\frac{1}{2}$				
x_1	$-\frac{1}{3}$			$-\frac{9}{5}$	
		$\frac{3}{2}$			$\frac{18}{5}$
x_2	0	1	$-\frac{9}{2}$		
		$-\frac{3}{2}$		$\frac{9}{5}$	
x_3	$\frac{1}{3}$				
x_4	$\frac{1}{2}$				

$$P(x) = 1 + 0 - \frac{9}{2}x^2 + 0 + \frac{18}{5}x^2\left(x^2 - \frac{1}{9}\right)$$

$$\frac{1}{2}\left(\frac{3}{2} - \frac{3}{2}\right)(x-0)$$

$$-\frac{9}{2}(x-0)\left(x - \frac{\frac{1}{3} - \frac{1}{3}}{2}\right)$$